Bench Press - Linear Regression Model - Peak Power

Model Summary - Peak Power T2

Model	R	R^2	Adjusted R^2	RMSE	Durbin-Watson		
					Autocorrelation	Statistic	p
M_0	0.000	0.000	0.000	38.23	−0.108	1.738	.570
M_1	0.663	0.439	0.404	29.51	0.274	1.046	.026
M_2	0.768	0.589	0.534	26.09	0.015	1.604	.343

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	10906	1	10905.8	12.52	.003
	Residual	13936	16	871.0		
	Total	24842	17			
M_2	Regression	14635	2	7317.6	10.75	.001
	Residual	10207	15	680.4		
	Total	24842	17			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p	95% CI		Collinearity Statistics	
							Lower	Upper	Tolerance	VIF
M_0	(Intercept)	57.23	9.010		6.352	< .001	38.22	76.24		
M_1	(Intercept)	181.13	35.700		5.074	< .001	105.45	256.81		
	Co-contraction duration_T7	−163.64	46.246	−0.663	−3.539	.003	−261.68	−65.60	1.000	1.000
M_2	(Intercept)	132.83	37.701		3.523	.003	52.47	213.19		
	Co-contraction duration_T7	−121.29	44.698	−0.491	−2.714	.016	−216.57	−26.02	0.836	1.196
	Delta Onset_T7	−112.88	48.217	−0.424	−2.341	.033	−215.65	−10.11	0.836	1.196

Note. The following covariates were considered but not included: Cycle Duration_T7, Delta Peak-Onset Ag (s)_T7, Abs Delta Onset %Cycle_T7, Motion EMD Ant_T7, Co-Contraction/ Cycle Duration_T7, Delta Peak-Onset Ant (s)_T7, Motion EMD Ant %Cycle_T7.

Part And Partial Correlations

Model		Partial	Part
M_1	Co-contraction duration_T7	−0.663	−0.663
M_2	Co-contraction duration_T7	−0.574	−0.449
	Delta Onset_T7	−0.517	−0.387

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Intercept)	Co-contraction duration_T7	Delta Onset_T7
M_1	1	1.981	1.000	0.01	0.01	
	2	0.019	10.166	0.99	0.99	
M_2	1	2.575	1.000	0.004	0.004	0.043
	2	0.411	2.505	0.006	0.015	0.707
	3	0.015	13.304	0.991	0.98	0.249

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	6.722	112.904	57.230	29.341	18
Residual	−43.000	61.036	2.765×10^{-15}	24.503	18
Std. Predicted Value	−1.721	1.897	8.603×10^{-17}	1.000	18
Std. Residual	−1.744	2.573	7.661×10^{-4}	1.015	18

Bench Press - Linear Regression Model W/ Sex) - Peak Power

Model Summary - Peak Power T2

					Durbin-Watson		
Model	R	R^2	Adjusted R^2	RMSE	Autocorrelation	Statistic	p
M_0	0.000	0.000	0.000	38.23	−0.108	1.738	.570
M_1	0.750	0.563	0.536	26.05	−0.101	1.770	.623
M_2	0.846	0.715	0.677	21.71	0.037	1.583	.341

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	13983	1	13983.2	20.60	< .001
	Residual	10859	16	678.7		
	Total	24842	17			
M_2	Regression	17770	2	8884.9	18.84	< .001
	Residual	7072	15	471.5		
	Total	24842	17			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

							95% CI		Collinearity Statistics	
Model		Unstandardized	Standard Error	Standardized[a]	t	p	Lower	Upper	Tolerance	VIF
M_0	(Intercept)	57.23	9.010		6.352	< .001	38.22	76.24		
M_1	(Intercept)	92.17	9.846		9.361	< .001	71.30	113.04		
	Gender (W)	−57.17	12.596		−4.539	< .001	−83.87	−30.47	1.000	1.000
M_2	(Intercept)	158.58	24.828		6.387	< .001	105.66	211.50		
	Gender (W)	−49.69	10.825		−4.590	< .001	−72.76	−26.62	0.941	1.063
	Cycle Duration_T7	−99.84	35.229	−0.403	−2.834	.013	−174.93	−24.75	0.941	1.063

Note. The following covariates were considered but not included: Delta Onset_T7, Co-contraction duration_T7, Delta Peak-Onset Ag (s)_T7, Abs Delta Onset %Cycle_T7, Motion EMD Ant_T7, Co-Contraction/ Cycle Duration_T7, Delta Peak-Onset Ant (s)_T7, Motion EMD Ant %Cycle_T7.

[a] Standardized coefficients can only be computed for continuous predictors.

Part And Partial Correlations

Model		Partial	Part
M_1	Gender	0.750	0.750
M_2	Gender	0.764	0.632
	Cycle Duration_T7	−0.591	−0.390

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

				Variance Proportions		
Model	Dimension	Eigenvalue	Condition Index	(Intercept)	Gender (W)	Cycle Duration_T7
M_1	1	1.782	1.000	0.109	0.109	
	2	0.218	2.857	0.891	0.891	
M_2	1	2.708	1.000	0.005	0.04	0.005
	2	0.271	3.160	0.026	0.944	0.019
	3	0.021	11.308	0.968	0.016	0.975

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	6.651	109.256	57.230	32.331	18
Residual	−35.540	48.432	3.356×10^{-15}	20.396	18
Std. Predicted Value	−1.564	1.609	-7.823×10^{-17}	1.000	18
Std. Residual	−1.874	2.525	0.001	1.043	18

Bench Press - Linear Regression Model - Relative Peak Power (W/kg)

Model Summary - PP (W/kg) T2

Model	R	R^2	Adjusted R^2	RMSE	Durbin-Watson		
					Autocorrelation	Statistic	p
M_0	0.000	0.000	0.000	0.468	−0.157	1.892	.815
M_1	0.668	0.446	0.411	0.359	0.155	1.343	.143

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	1.662	1	1.662	12.87	.002
	Residual	2.066	16	0.129		
	Total	3.727	17			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p	95% CI		Collinearity Statistics	
							Lower	Upper	Tolerance	VIF
M_0	(Intercept)	0.799	0.110		7.244	< .001	0.567	1.032		
M_1	(Intercept)	2.329	0.435		5.358	< .001	1.407	3.250		
	Co-contraction duration_T7	−2.020	0.563	−0.668	−3.587	.002	−3.213	−0.826	1.000	1.000

Note. The following covariates were considered but not included: Delta Onset_T7, Massa FInal, Delta Peak-Onset Ag (s)_T7, Abs Delta Onset %Cycle_T7, Motion EMD Ant_T7, Co-Contraction/ Cycle Duration_T7, Delta Peak-Onset Ant (s)_T7, Motion EMD Ant %Cycle_T7.

Part And Partial Correlations

Model		Partial	Part
M_1	Co-contraction duration_T7	−0.668	−0.668

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions	
				(Intercept)	Co-contraction duration_T7
M_1	1	1.981	1.000	0.01	0.01
	2	0.019	10.166	0.99	0.99

Note. The intercept model is omitted, as no meaningful information can be shown.

Residuals Statistics

	Minimum	Maximum	Mean	SD	N
Predicted Value	0.184	1.258	0.799	0.313	18
Residual	−0.593	0.703	1.699×10^{-17}	0.349	18
Std. Predicted Value	−1.969	1.468	4.539×10^{-17}	1.000	18
Std. Residual	−1.723	2.144	0.010	1.024	18

Bench Press - Linear Regression Model (w/ Sex) - Relative Peak Power (W/kg)

Model Summary - PP (W/kg) T2

					Durbin-Watson		
Model	R	R^2	Adjusted R^2	RMSE	Autocorrelation	Statistic	p
M_0	0.000	0.000	0.000	0.468	−0.157	1.892	.815
M_1	0.668	0.446	0.411	0.359	0.155	1.343	.143

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	1.662	1	1.662	12.87	.002
	Residual	2.066	16	0.129		
	Total	3.727	17			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

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Model		Unstandardized	Standard Error	Standardized[a]	t	p	Lower	Upper	Tolerance	VIF
M_0	(Intercept)	0.799	0.110		7.244	< .001	0.567	1.032		
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Note. The following covariates were considered but not included: Delta Onset_T7, Massa FInal, Delta Peak-Onset Ag (s)_T7, Abs Delta Onset %Cycle_T7, Motion EMD Ant_T7, Co-Contraction/ Cycle Duration_T7, Delta Peak-Onset Ant (s)_T7, Motion EMD Ant %Cycle_T7.

[a] Standardized coefficients can only be computed for continuous predictors.

Part And Partial Correlations

Model		Partial	Part
M_1	Co-contraction duration_T7	−0.668	−0.668

Note. The intercept model is omitted, as no meaningful information can be shown.

Collinearity Diagnostics

				Variance Proportions	
Model	Dimension	Eigenvalue	Condition Index	(Intercept)	Co-contraction duration_T7
M_1	1	1.981	1.000	0.01	0.01
	2	0.019	10.166	0.99	0.99

Note. The intercept model is omitted, as no meaningful information can be shown.

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